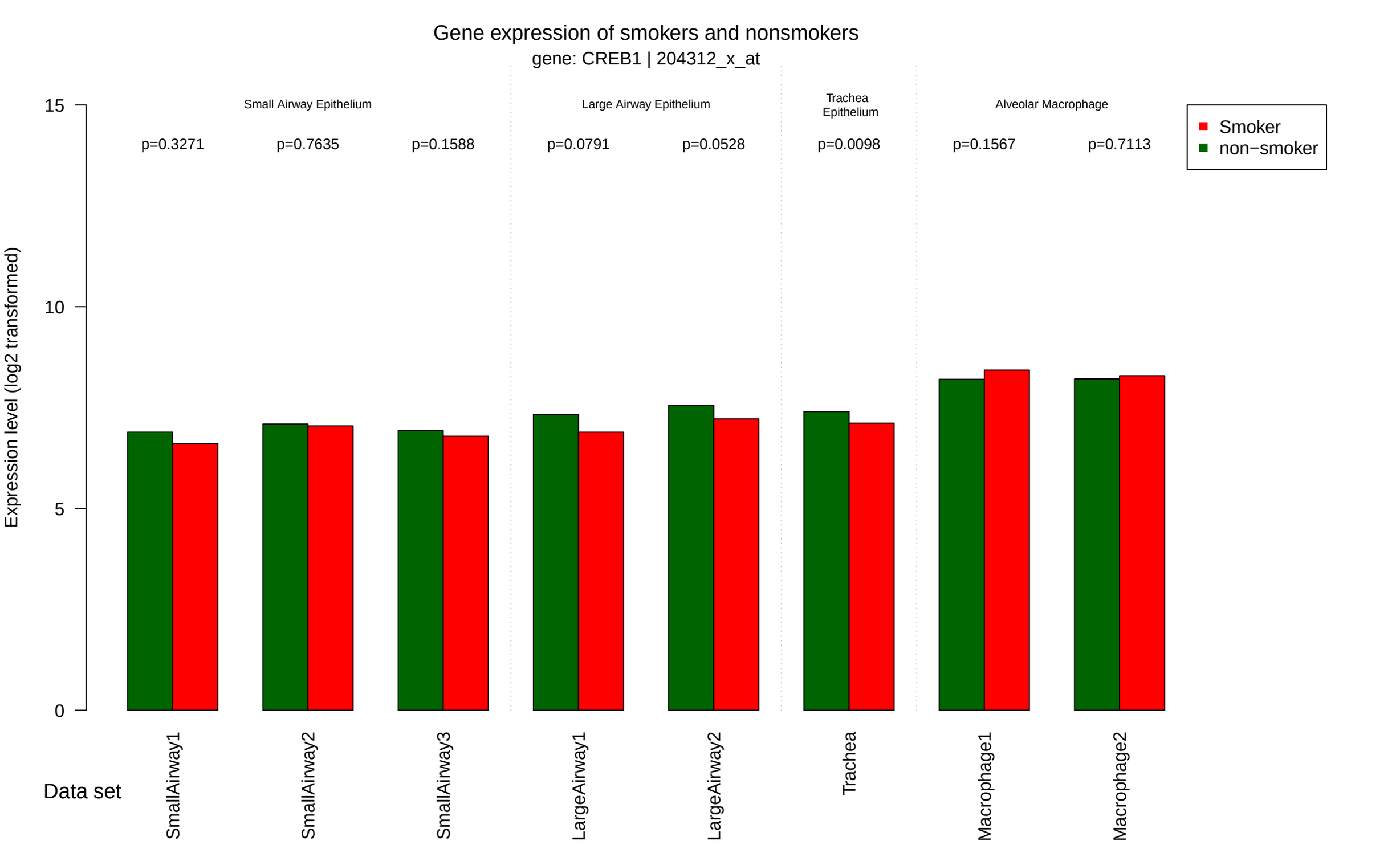
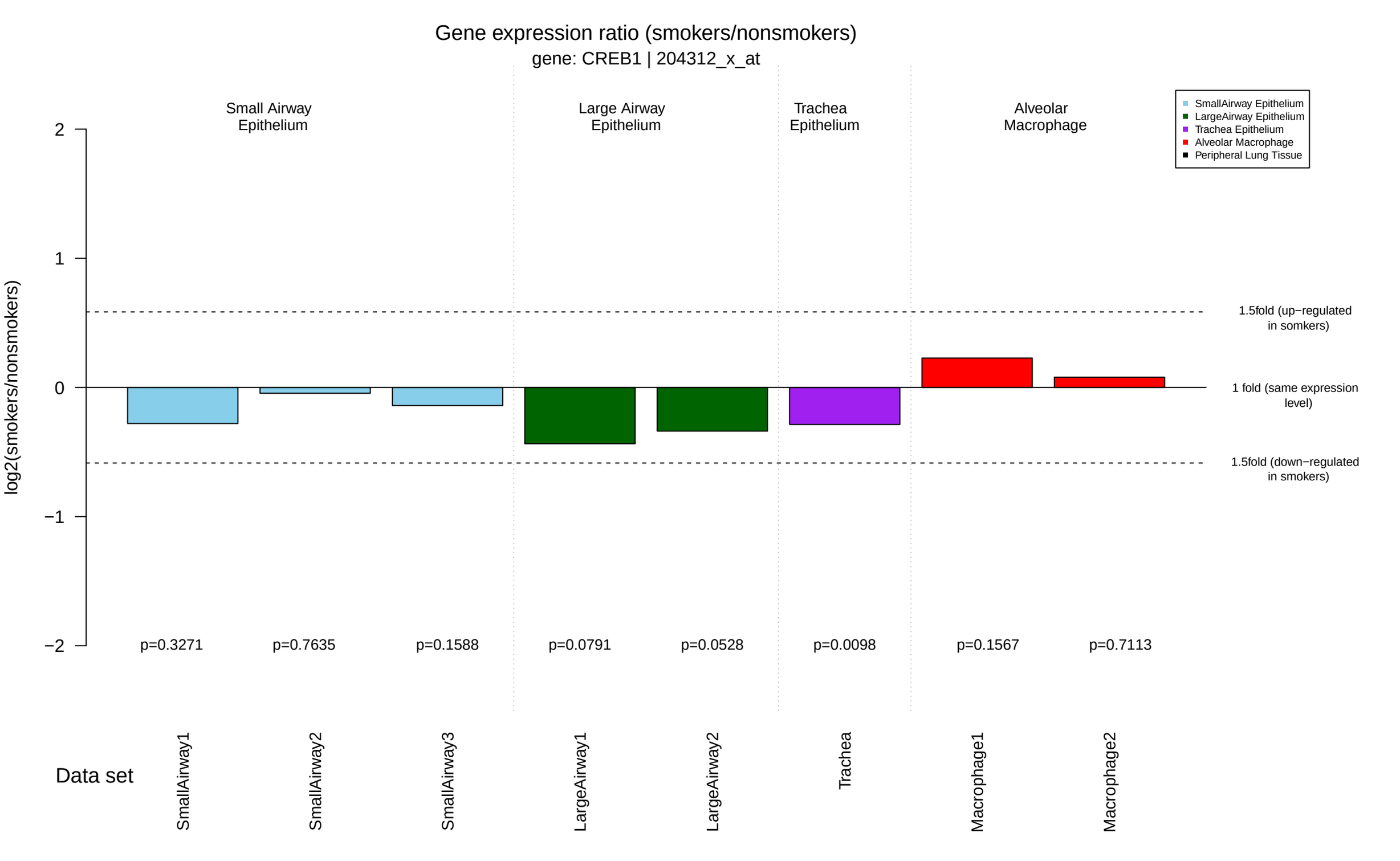
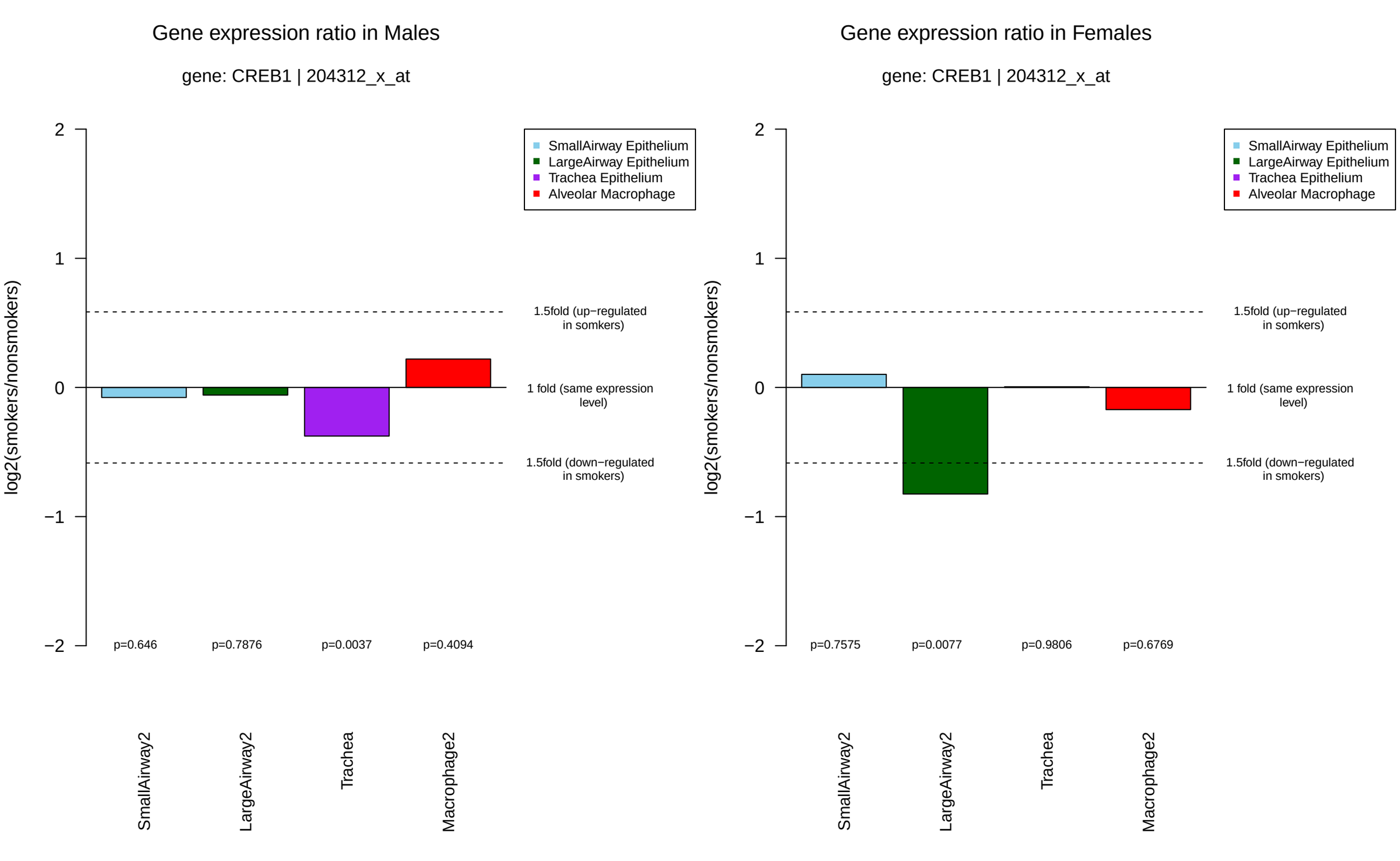
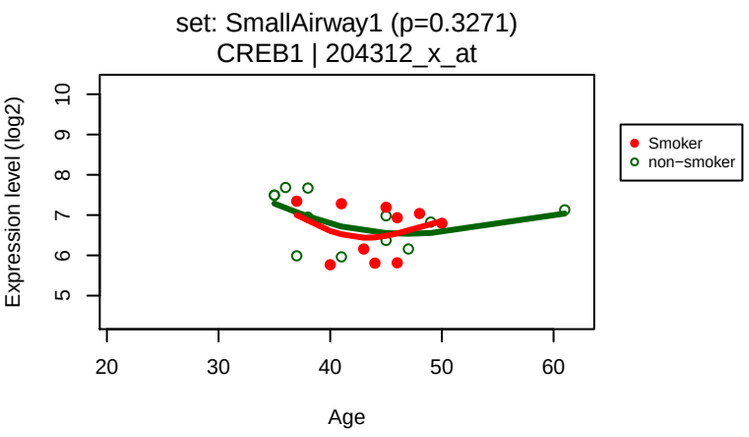
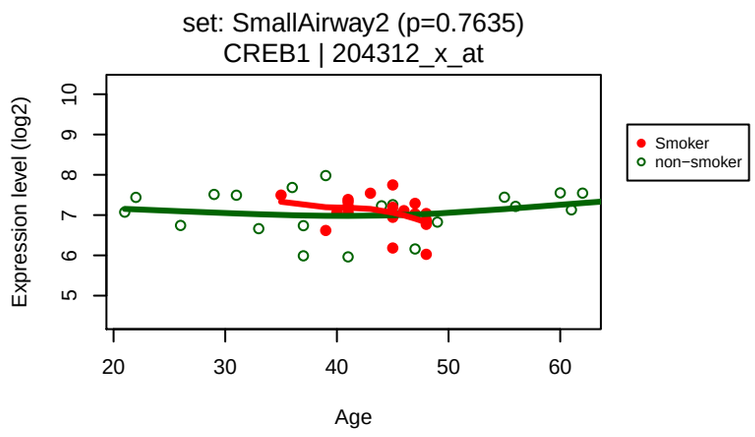
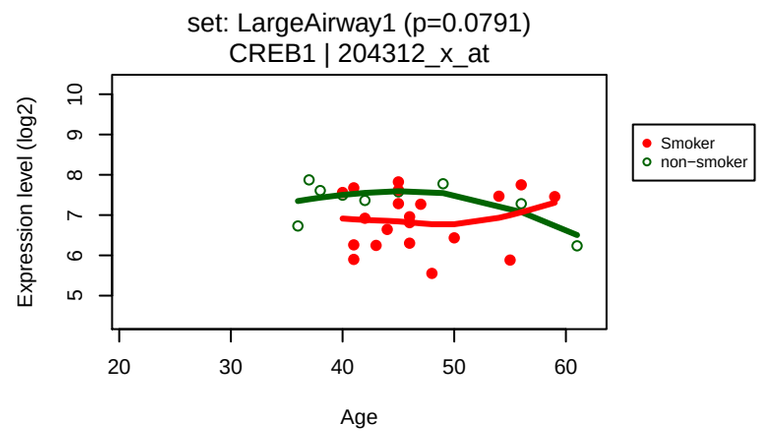

Gene expression of smokers and nonsmokers
gene: CREB1 | 204312_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3271
p=0.7635
p=0.1588
p=0.0791
p=0.0528
p=0.0098
p=0.1567
p=0.7113
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

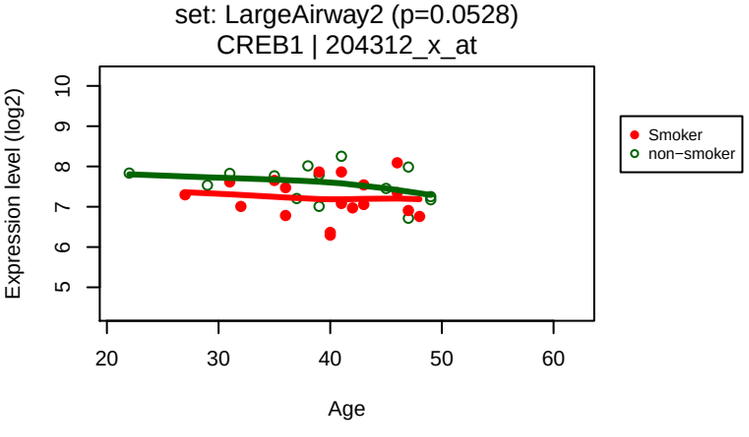

Gene expression ratio (smokers/nonsmokers)
gene: CREB1 | 204312_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3271
p=0.7635
p=0.1588
p=0.0791
p=0.0528
p=0.0098
p=0.1567
p=0.7113
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

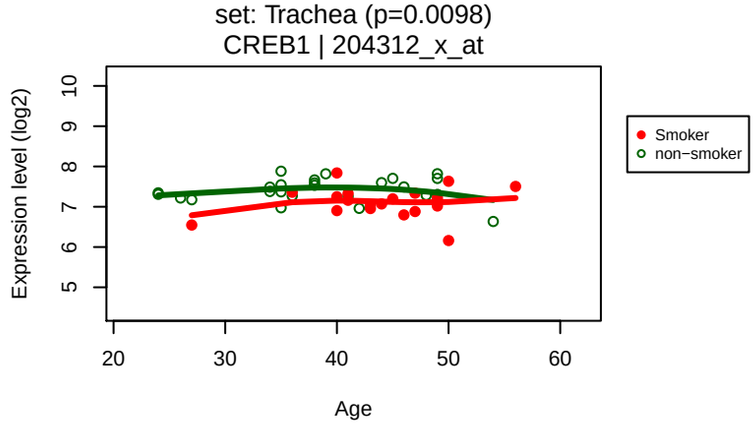

Gene expression ratio in Males
gene: CREB1 | 204312_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.646
p=0.7876
p=0.0037
p=0.4094
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CREB1 | 204312_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7575
p=0.0077
p=0.9806
p=0.6769
SmallAirway2
LargeAirway2
Trachea
Macrophage2

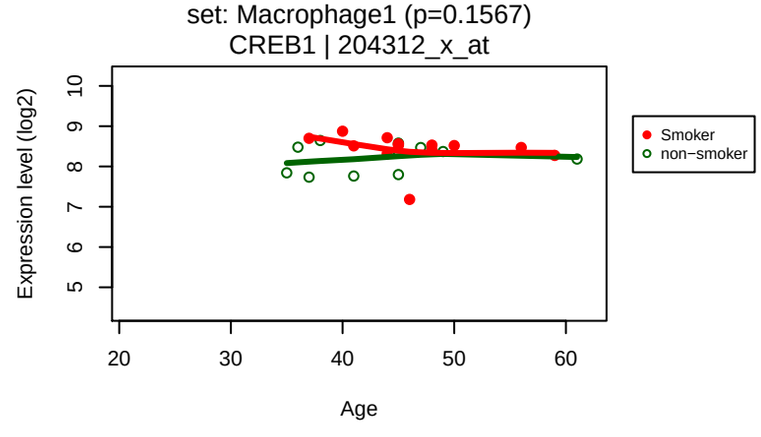

set: SmallAirway1 (p=0.3271)
CREB1 | 204312_x_at
Expression level (log2)
Age
Smoker
non−smoker

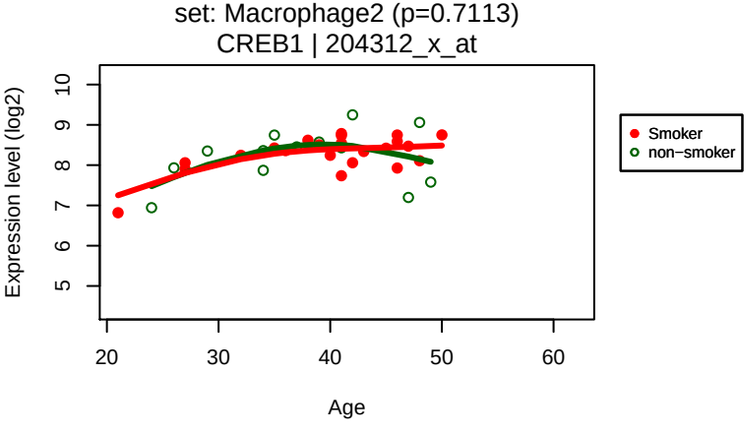

set: SmallAirway2 (p=0.7635)
CREB1 | 204312_x_at
Expression level (log2)
Age
Smoker
non−smoker

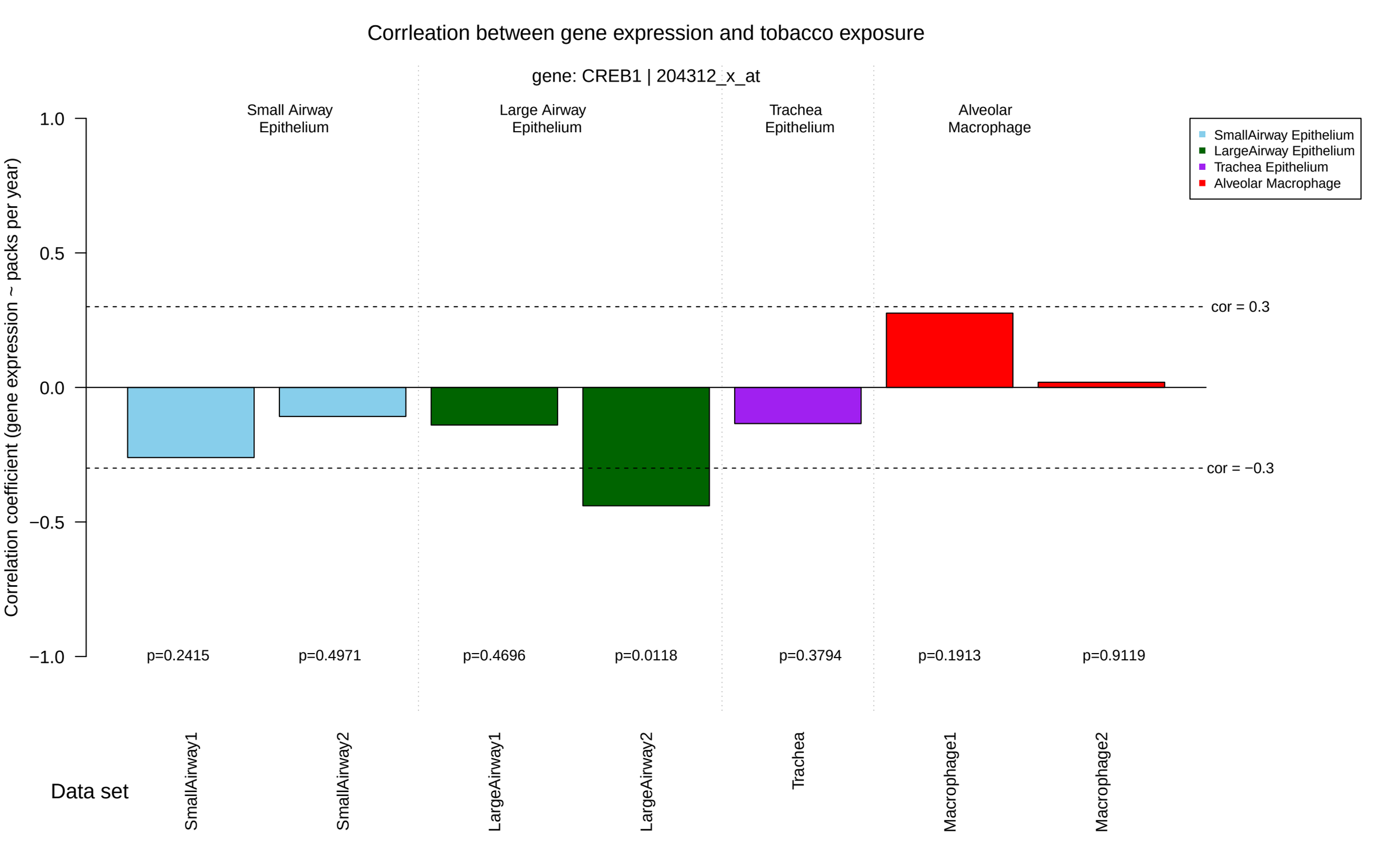

set: LargeAirway1 (p=0.0791)
CREB1 | 204312_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0528)
CREB1 | 204312_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0098)
CREB1 | 204312_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1567)
CREB1 | 204312_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7113)
CREB1 | 204312_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CREB1 | 204312_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2415
p=0.4971
p=0.4696
p=0.0118
p=0.3794
p=0.1913
p=0.9119
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2